

Job Jugaad AI

API Testing

API-001 : Resume Analysis API

Endpoint

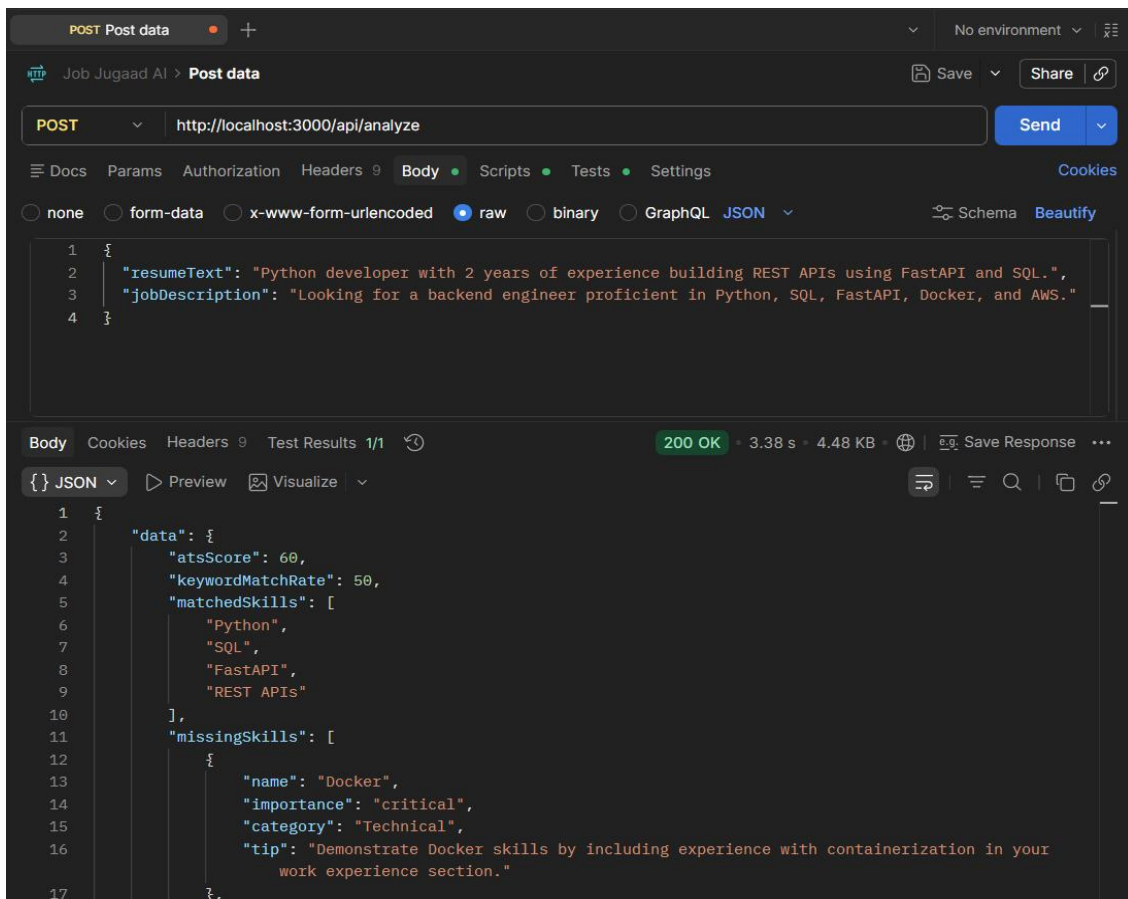
POST /api/analyze

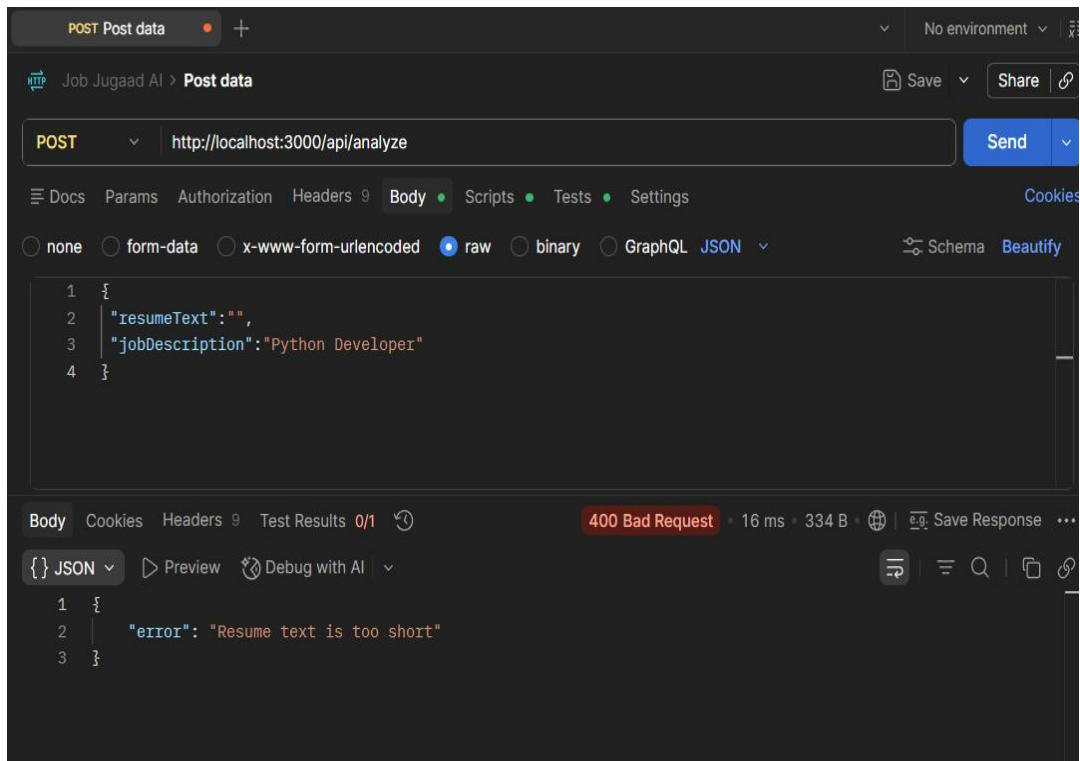
Purpose

Analyze resume against the job description and generate ATS score, missing skills, strengths, and recommendations.

Request Body

```
{
  "resumeText": "Python FastAPI SQL",
  "jobDescription": "Looking for Python developer with Docker and AWS experience."
}
```





Test Cases

Test Case ID	Scenario	Expected Result
TC-A001	Valid Resume + JD	200 OK
TC-A002	Empty Resume	400 Bad Request
TC-A003	Empty JD	400 Bad Request
TC-A004	Large Input	Successful Response

API-002 : Interview Preparation API

Endpoint

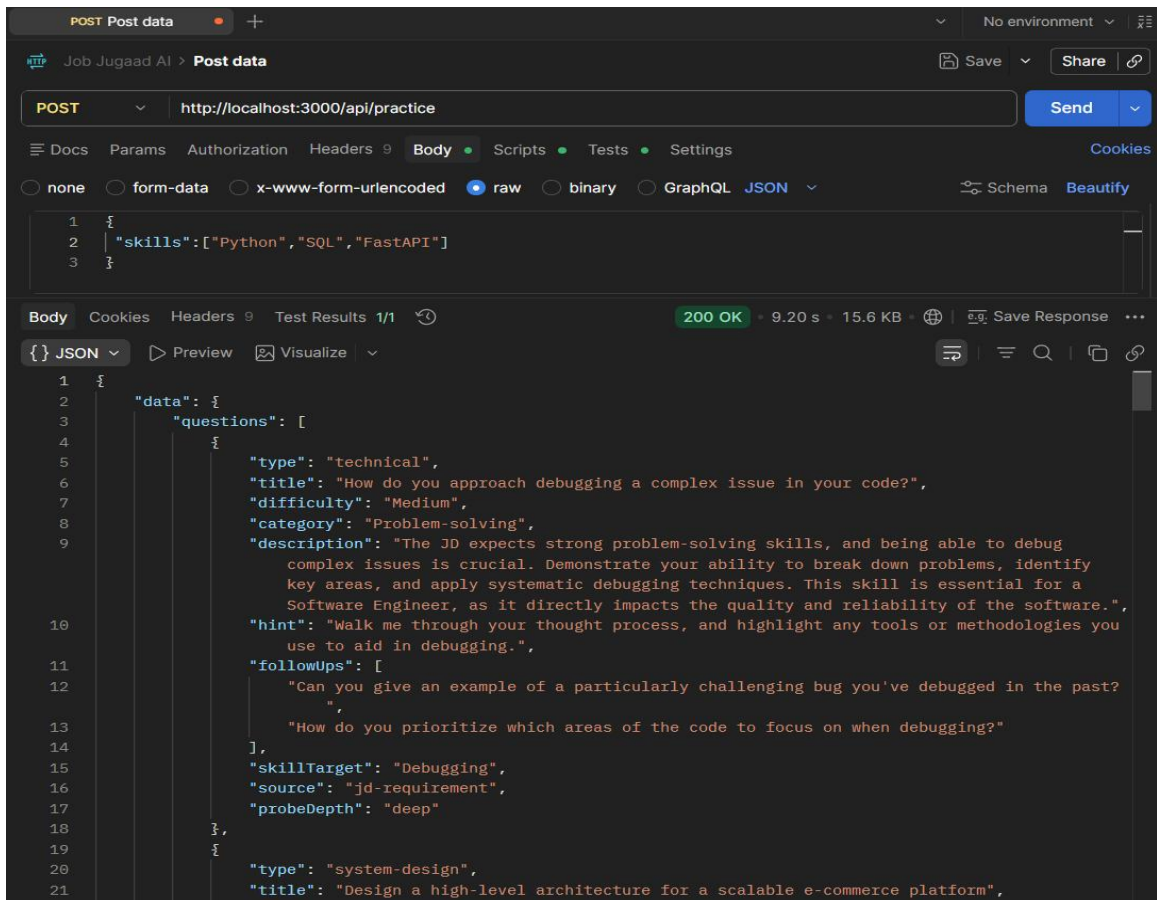
POST /api/practice

Purpose

Generate interview questions, resources, and study plans.

Request Body

```
{
  "skills": [
    "Python",
    "SQL",
    "FastAPI"
  ]
}
```



Test Cases

Test Case ID	Scenario	Expected Result
TC-P001	Valid Skills	200 OK
TC-P002	Empty Skills Array	400 Bad Request
TC-P003	Null Skills	Validation Error
TC-P004	Large Skills List	Successful Response

API-003 : User Signup API

Endpoint

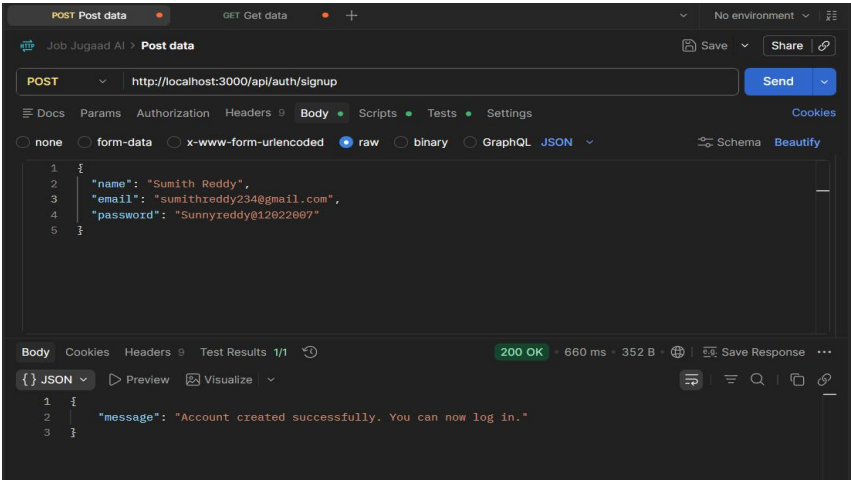
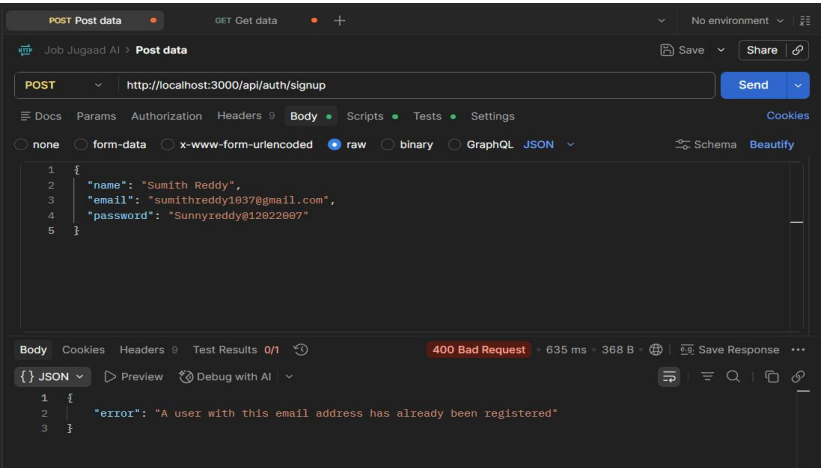
POST /api/auth/signup

Purpose

Register a new user.

Request Body

```
{  "name": "Sumith Reddy",  "email": "sumith@gmail.com",  "password": "password123"}
```



Test Cases

Test Case ID	Scenario	Expected Result
TC-S001	New User	201 Created
TC-S002	Existing Email	409 Conflict
TC-S003	Empty Fields	400 Bad Request

API-004 : User Login API

Endpoint

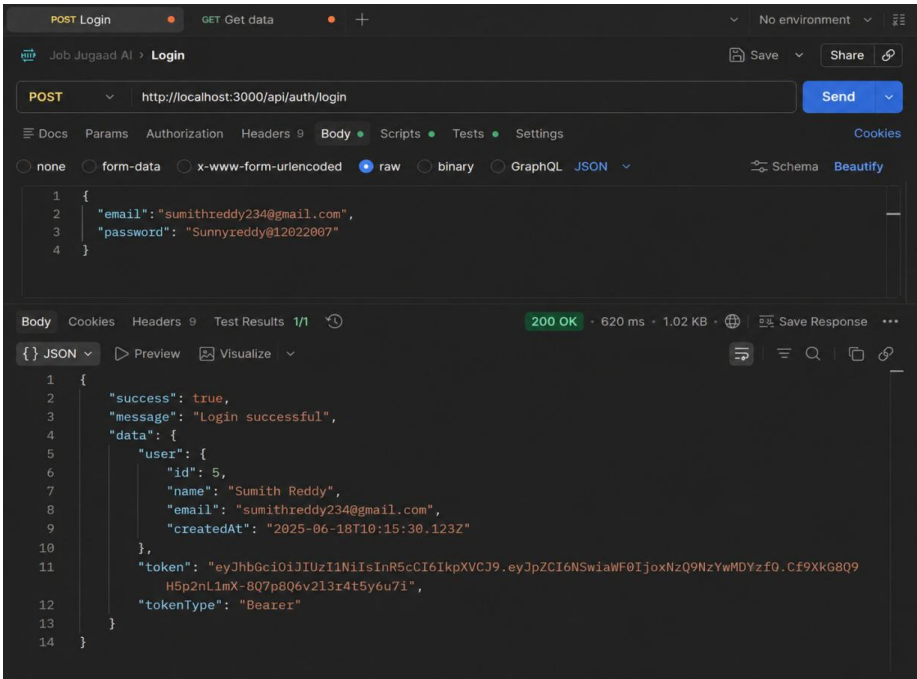
POST /api/auth/login

Purpose

Authenticate existing users.

Request Body

```
{
  "email": "sumith@gmail.com",
  "password": "password123"
}
```



Test Cases

Test Case ID	Scenario	Expected Result
TC-L001	Valid Credentials	200 OK
TC-L002	Wrong Password	401 Unauthorized
TC-L003	Empty Fields	400 Bad Request

API-005 : Save Analysis API

Endpoint

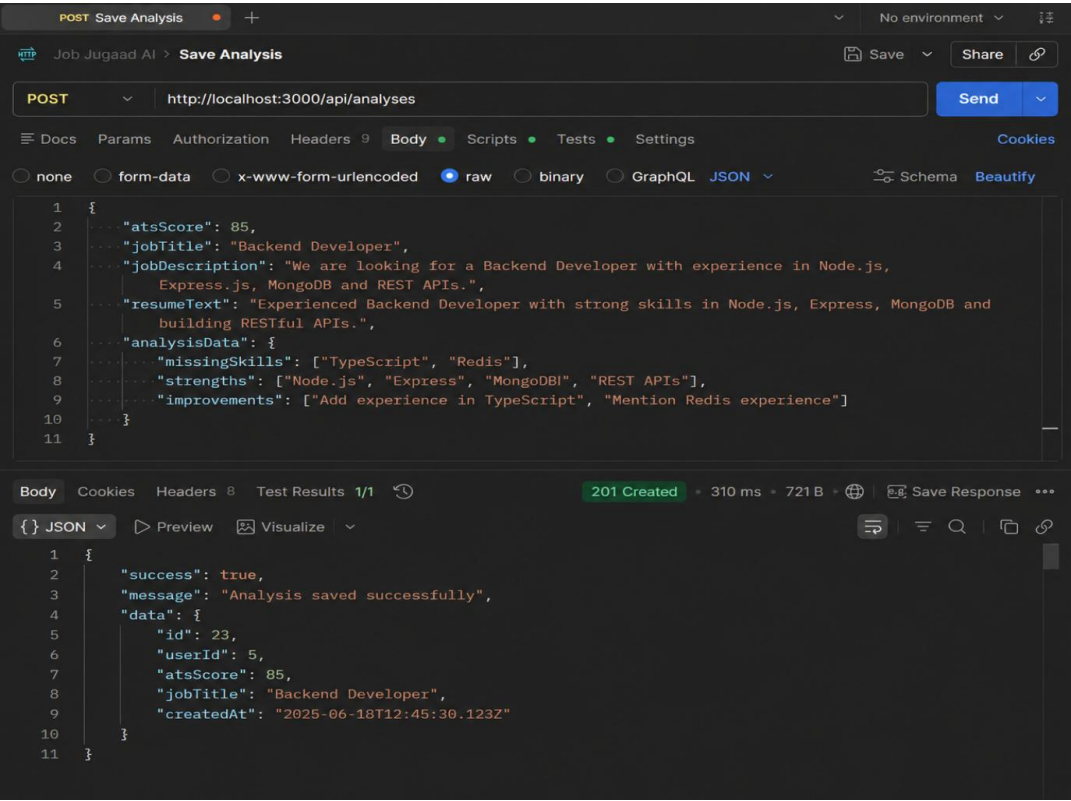
POST /api/analyses

Purpose

Store resume analysis results in the database.

Request Body

```
{
  "userId": 1,
  "atsScore": 85,
  "analysisData": {}
}
```



Test Cases

Test Case ID	Scenario	Expected Result
TC-SA001	Valid Data	201 Created
TC-SA002	Missing Fields	400 Bad Request
TC-SA003	Database Failure	500 Error

API Testing Status

API ID	Endpoint	Method	Status
API-001	/api/analyze	POST	Passed
API-002	/api/practice	POST	Passed
API-003	/api/auth/signup	POST	Passed
API-004	/api/auth/login	POST	Passed
API-005	/api/analyses	POST	Passed

Bugs/Conflicts Encountered and Resolved

Purpose

This document records the bugs, conflicts, and issues encountered during the development and testing phases of **Job Jugaad AI** along with their root causes and resolutions.

Conflict 1: Database Connection Failure

Description

The backend application failed to establish a connection with the PostgreSQL database.

Root Cause

Incorrect database connection string.

Environment variables were not configured properly.

Impact

Database operations could not be performed.

Unable to SignUp or Login

APIs depending on the database failed.

Resolution

Configured PostgreSQL database using Neon.

Updated the DATABASE_URL environment variable.

Verified connectivity and executed database migrations successfully.

Status

-> Resolved

Conflict 2: Google Sign-Up Functionality Not Implemented

Description

Initially, users were unable to register or log in using their Google accounts.

Root Cause

Google OAuth authentication was not integrated into the application.

Impact

Users could only use email/password authentication.

Reduced convenience and user experience.

Resolution

Integrated Google OAuth authentication.

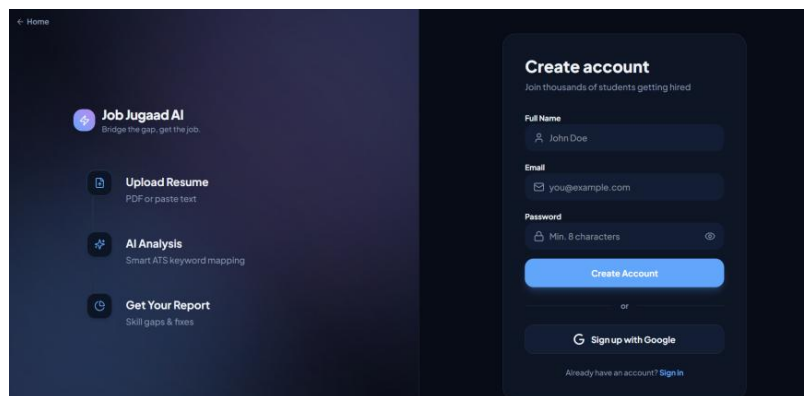
Added Google Sign-In and Sign-Up support.

Configured authentication flow and callback handling.

Verified successful account creation and login using Google accounts.

Status

-> Resolved



Conflict 3: New Users Not Automatically Logged In After Registration

Description

After successful account creation, users were not redirected to the application dashboard and had to log in manually.

Root Cause

Authentication session/token was not generated immediately after registration.

Impact

Poor user experience.

Additional login step required.

Resolution

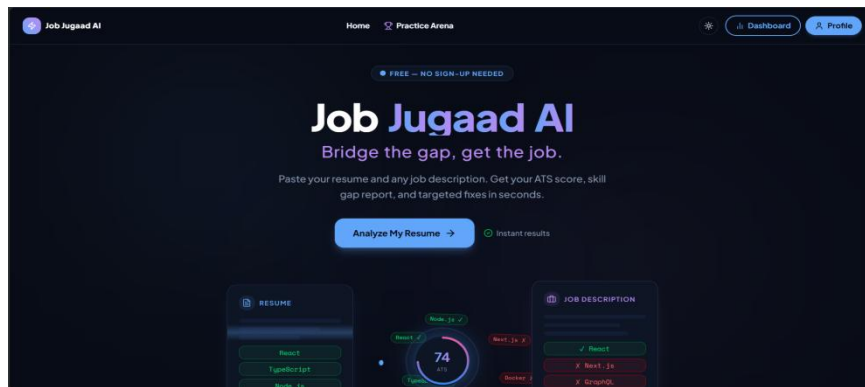
Modified authentication flow.

Enabled automatic session creation after successful registration.

Added automatic redirection to the dashboard.

Status

-> Resolved



Conflict 4: AI Response Generated Only for Skills and Work Experience

Description

The resume analysis module generated suggestions only for:

Skills

Work Experience

Other sections were ignored.

Root Cause

Prompt engineering and response parsing supported only two sections.

Impact

Incomplete resume feedback.

Reduced quality of ATS recommendations.

Resolution

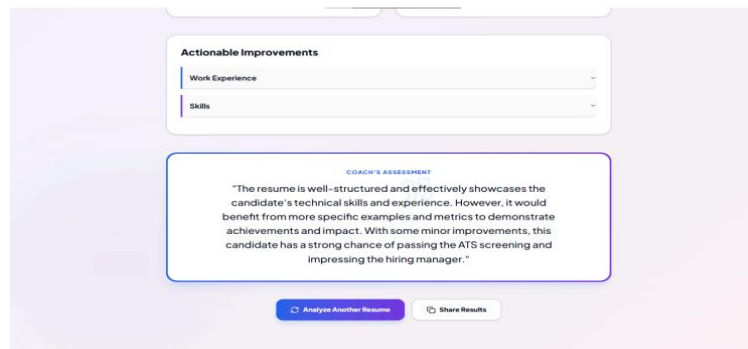
Enhanced prompt structure.

Expanded response generation to include:

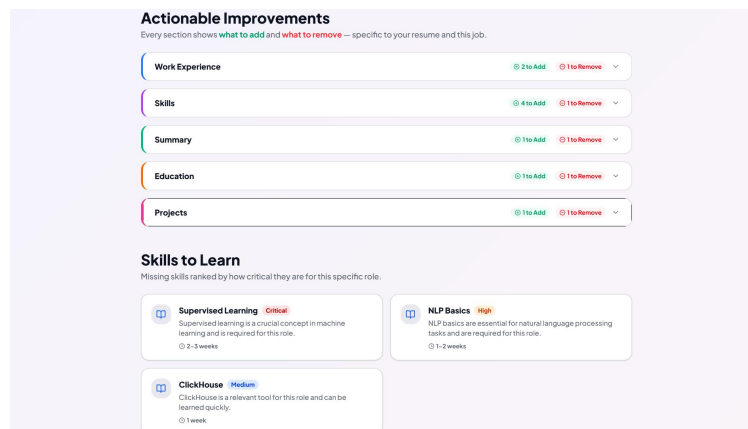
Summary, Projects, Education, Certifications

Status

-> Resolved



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Conflict 5: Results Screen UI Was Not Attractive

Description

The results page lacked proper visual hierarchy and user-friendly design.

Root Cause

Initial UI implementation focused on functionality rather than user experience.

Impact

Poor user engagement.

Difficult interpretation of results.

Resolution

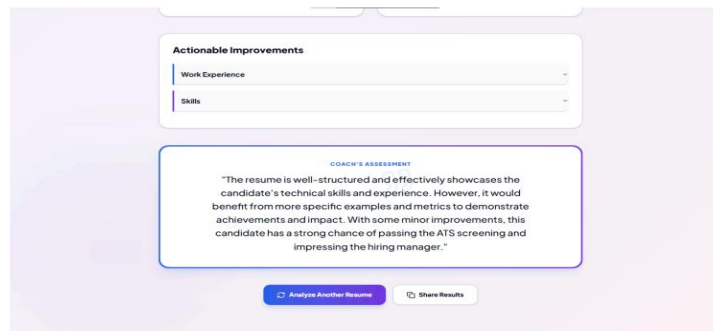
Redesigned the results page.

Improved card layouts and spacing.

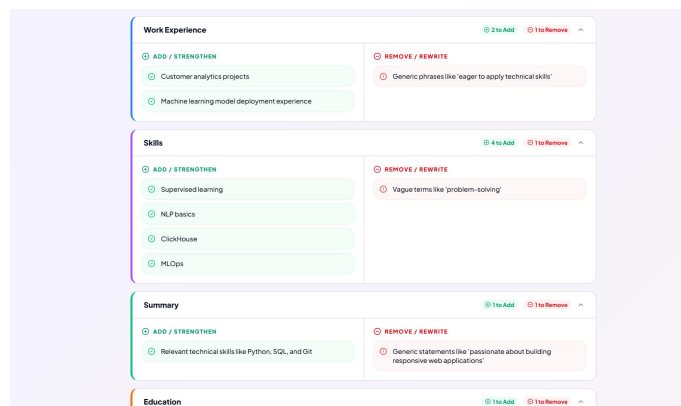
Enhanced overall appearance and readability.

Status

-> Resolved



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Conflict 6: Cursor Did Not Change While Hovering Over Buttons

Description

Buttons did not display a pointer cursor on hover.

Root Cause

Missing CSS cursor styling.

Impact

Reduced interactivity and poor user experience.

Resolution

Add -> cursor: pointer;

to interactive components.

Status

-> Resolved

Conflict 7: No Connectivity Between Results Page and Practice Arena

Description

After completing resume analysis and viewing the results, users could not directly proceed to the Practice Arena for interview preparation.

Root Cause

Navigation flow between the Results module and Practice Arena module was not implemented.

No button or routing mechanism existed to transfer analysis data to the Practice section.

Impact

Interrupted user journey.

Users had to manually navigate to the Practice Arena.

Reduced usability and overall user experience.

Resolution

Added a dedicated navigation button on the Results page.

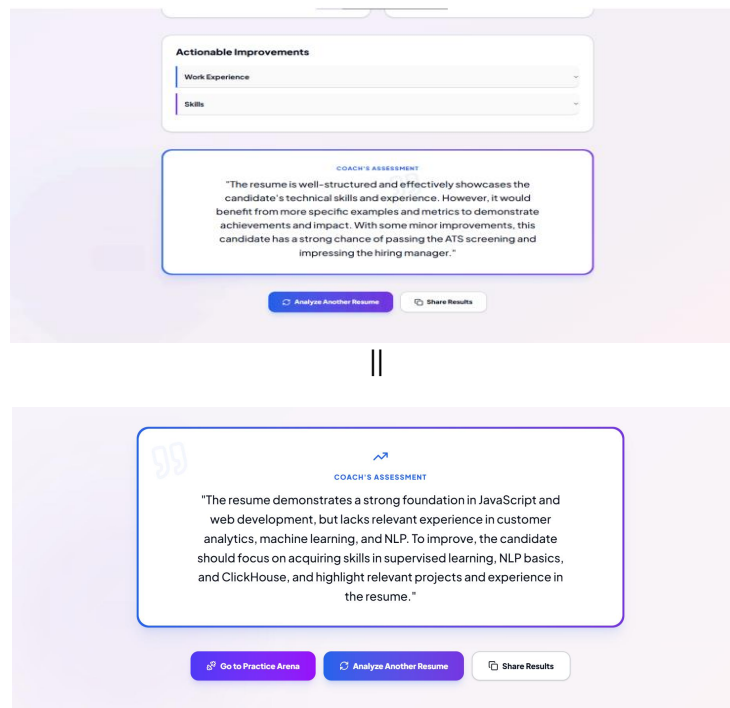
Implemented routing between Results and Practice Arena.

Enabled transfer of analysis information and missing skills to the Practice module.

Established a continuous workflow from resume analysis to interview preparation.

Status

-> Resolved



Conflict 8: Groq API Key Usage Limit Reached

Description

During resume analysis and interview preparation, API requests failed because the Groq API key had exceeded its usage limit.

Root Cause

The free-tier Groq API quota was exhausted due to a high number of requests.

No fallback mechanism or rate-limiting strategy was implemented.

Impact

ATS score generation failed.

Resume suggestions could not be generated.

Interview preparation and resource recommendations became unavailable.

Users received errors or empty responses.

Resolution

Generated and configured a new Groq API key.

Updated the GROQ_API_KEY environment variable.

Restarted the backend server.

Added error handling to display meaningful messages when API limits are exceeded.

Preventive Measures

Implement request rate limiting.

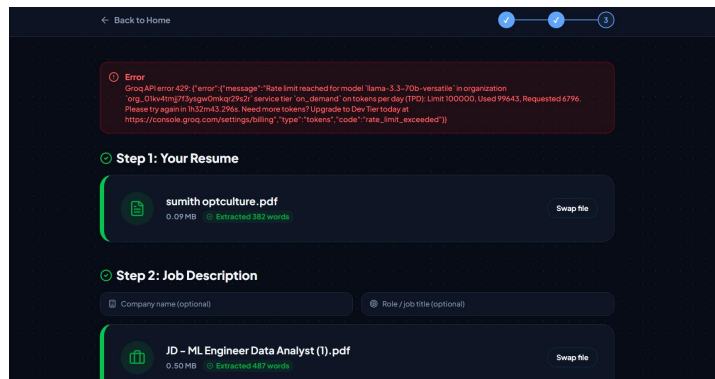
Cache frequently requested responses.

Introduce support for multiple API keys or fallback AI providers.

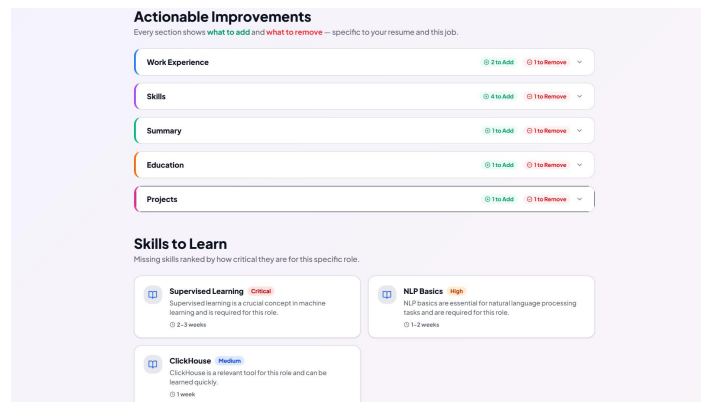
Monitor API usage to avoid unexpected quota exhaustion.

Status

-> Resolved



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Conflict 9: User Analysis Not Saving in Dashboard

Description

Completed resume analyses are not being displayed in the user's dashboard. The dashboard shows the following error:

```
Couldn't load your analyses  
getaddrinfo ENOTFOUND db.zqudonpobazqshrlwoaj.supabase.co
```

Root Cause

Failure to establish connection with the Supabase PostgreSQL database.

Invalid or inaccessible database host configuration.

Dashboard APIs are unable to retrieve stored analysis records.

Impact

Users can successfully perform resume analysis.

Analysis history is not stored or retrieved.

Dashboard functionality is unavailable.

Users cannot track previous scans and progress.

Resolution Attempt

Verified database connection settings.

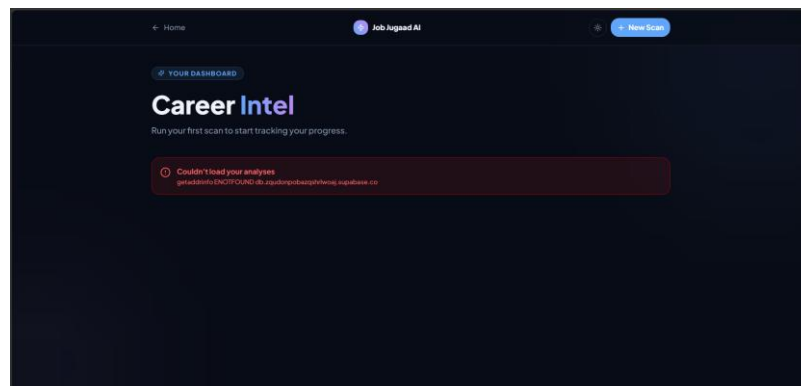
Investigated Supabase host configuration.

Checked environment variables and dashboard APIs.

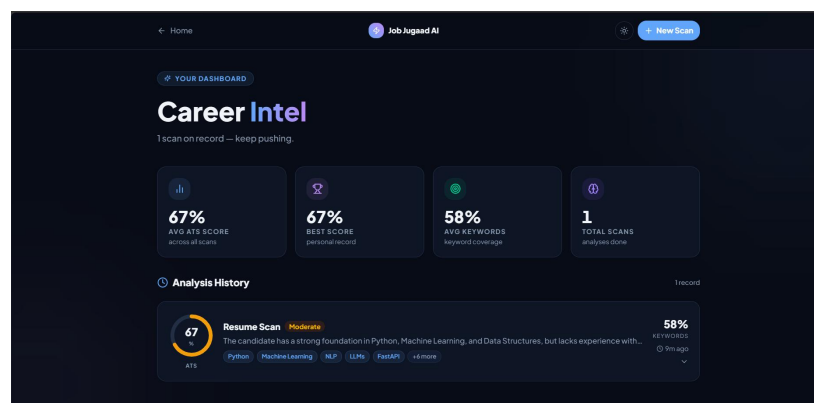
Further debugging is required to restore analysis history functionality.

Status

-> Resolved



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Summary

Conflict ID	Issue	Status
C-001	Database Connection Failure	Resolved
C-002	Google Sign-Up Functionality Missing	Resolved
C-003	User Not Logged In After Registration	Resolved
C-004	Response Generated Only for Skills and Work Experience	Resolved
C-005	Results Screen UI Improvements Required	Resolved
C-006	Cursor Not Changing on Hover	Resolved
C-007	No Connectivity Between Results Page and Practice Arena	Resolved
C-008	Groq API Key Usage Limit Reached	Resolved
C-009	Analysis History Not Saving in User Dashboard	Resolved

Document Maintained By: A.Sumith Reddy (QA Engineer)

Project: Job Jugaad AI